

SMART INDIA HACKATHON 2026



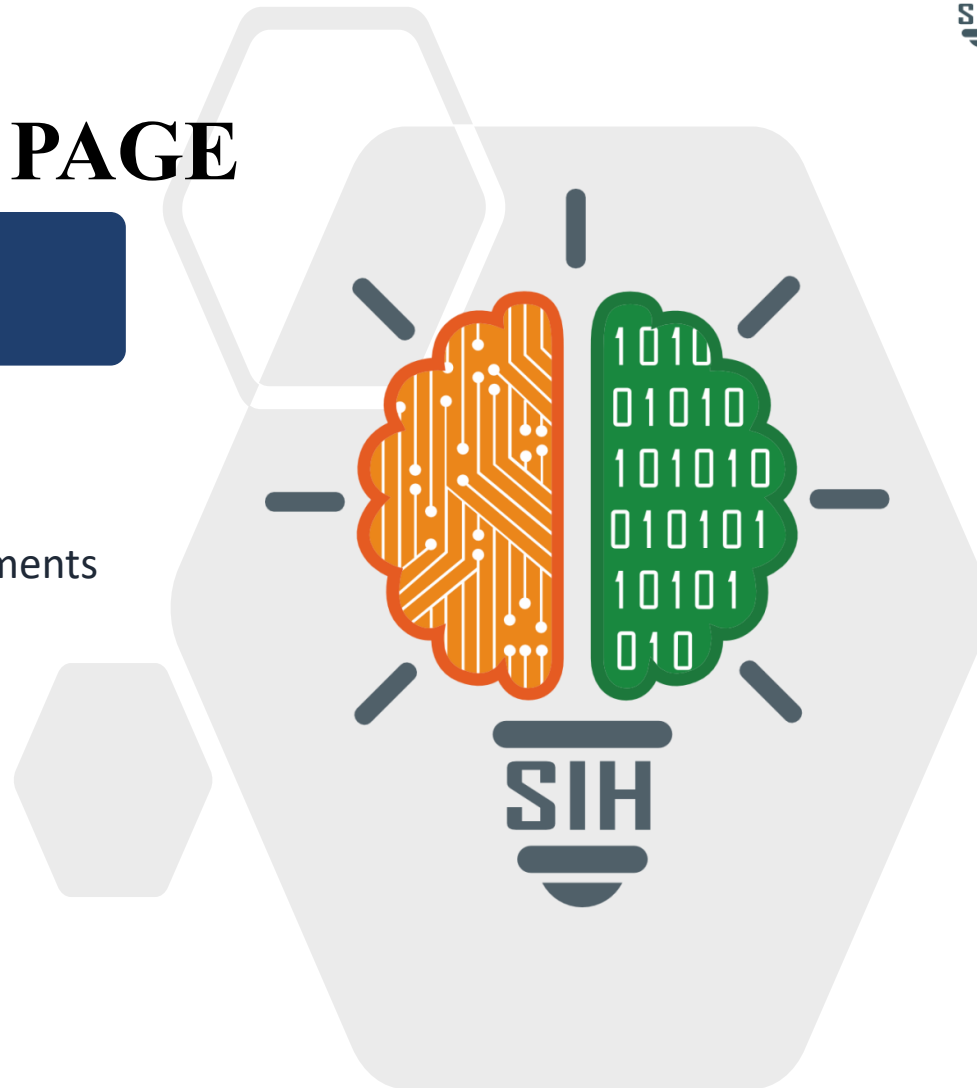
TITLE PAGE

TULA Unified Digital Lifecycle Platform for Legal Metrology

- **Problem Statement ID** – 26036
- **Problem Statement Title** – Development of an Online Verification System for Weighing and Measuring Instruments
- **Theme** – Miscellaneous
- **PS Category** – Software
- **Team ID** – _____
- **Team Name** – FriendlyFire

One instrument → one digital ID → one certificate anyone can check

Live prototype: tula-legal-metrology.vercel.app





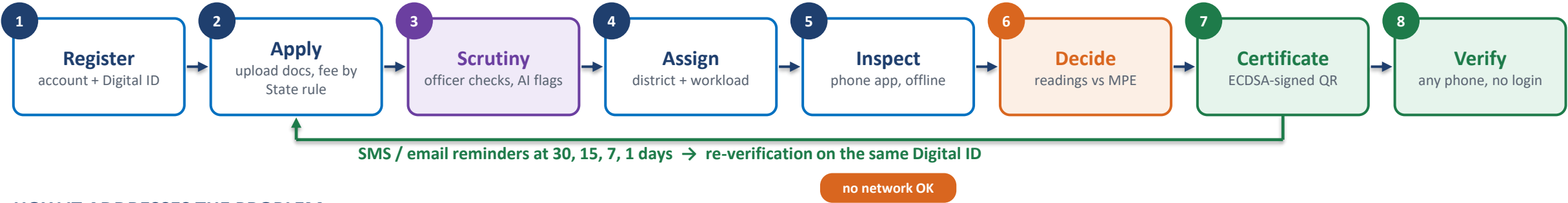
TULA: ONE INSTRUMENT, ONE DIGITAL ID



PROPOSED SOLUTION

A live web + mobile platform: every instrument gets a permanent Digital ID, and application, scrutiny, field inspection (works offline), signed QR certificate, reminders and enforcement all attach to it.

DETAILED EXPLANATION: the life of one instrument (built and running)



HOW IT ADDRESSES THE PROBLEM

Paper records in isolated systems
→ **One live database for every State**

Manual scheduling, pendency
→ **Enforced workflow + pendency dashboard**

Fake or copied certificates
→ **Signed QR: edited copies fail the scan**

Missed re-verification dates
→ **Automatic SMS / email reminders**

INNOVATION AND UNIQUENESS

Unfakeable QR
Server-signed (ECDSA P-256). Checked on any phone, even offline, plus a live revocation list

Offline field app
Installable app. Inspection saves on the phone and syncs, signs and notifies on reconnect

Computed verdict
Readings are checked against OIML MPE limits. PASS is impossible unless every check passes

Multi-State by design
Fees, validity and officers are data per State. Database rules isolate each State

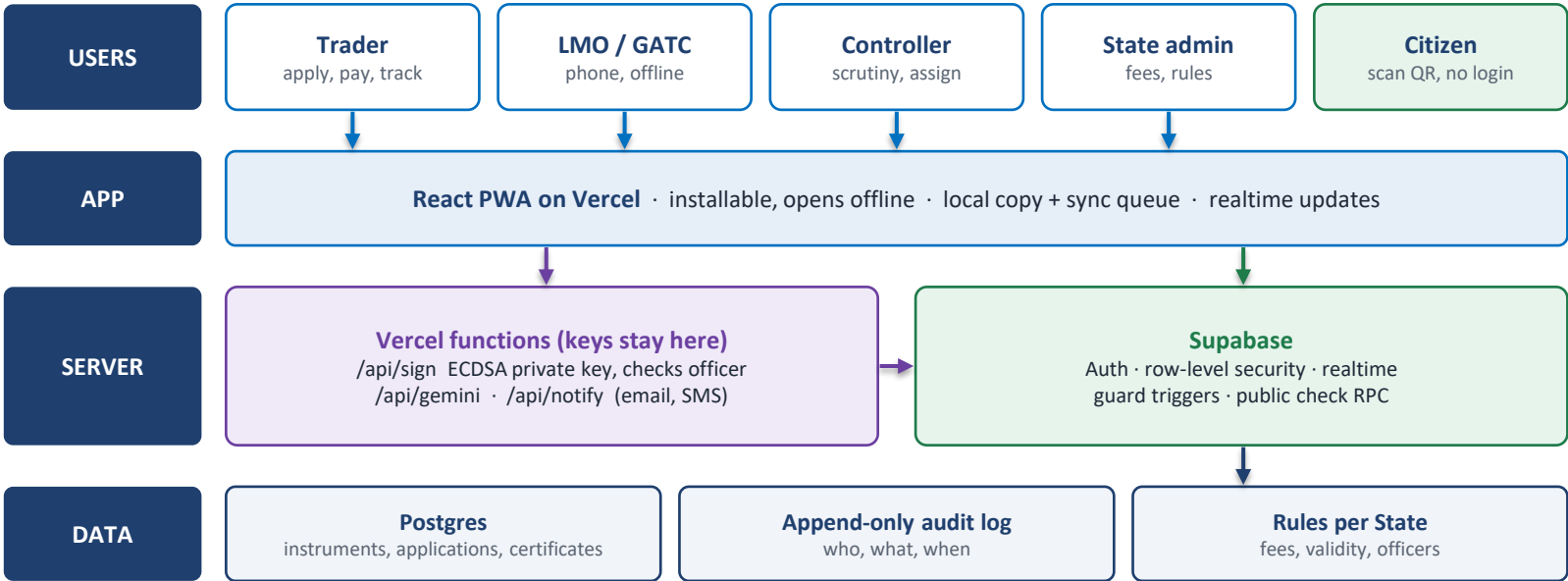
AI copilot, human decides
Gemini flags odd documents. It never approves, prices or passes anything



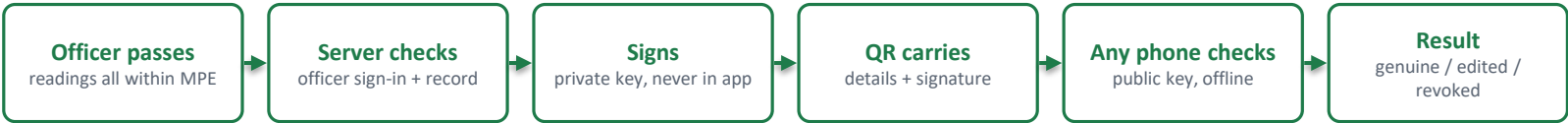
TECHNICAL APPROACH



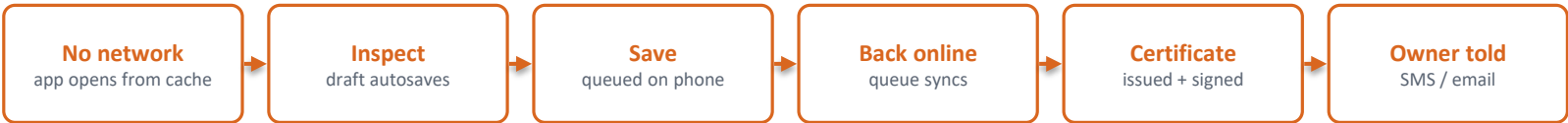
METHODOLOGY AND PROCESS FOR IMPLEMENTATION (as deployed)



HOW A FAKE QR IS CAUGHT



OFFLINE FIELD INSPECTION



TECHNOLOGIES USED

- React 19 · TypeScript · Vite · Tailwind role-based, mobile-first UI
- Service worker · local store · sync queue offline work, auto sync
- Supabase Postgres + RLS live data, isolated per State
- Supabase Auth · Realtime · RPC real accounts, live screens
- Vercel serverless functions signing, AI, notifications
- ECDSA P-256 · WebCrypto · jsQR sign, verify, real QR scan
- Google Gemini (Flash) document flags, help chat
- Resend · Twilio email and SMS updates
- jsPDF · QR · CSV certificate PDF, reports

IMPLEMENTATION PHASES

- 1 Done: live prototype, 6 roles, full lifecycle
- 2 Field trial with LMOs in one district
- 3 NIC SMS, treasury payment, e-Pramaan
- 4 State rollout on State cloud



ANALYSIS OF THE FEASIBILITY OF THE IDEA: what is built vs planned

PS requirement	Live today	Next
Registration, profiles, secure login	✓ real accounts, 6 roles	e-Pramaan, phone OTP
Apply for (re-)verification	✓ own instruments, docs, fee	treasury payment
Scheduling, allocation LMO / GATC	✓ district + workload	officer calendars
Inspection results recorded	✓ MPE computed, photos, GPS	cloud photo storage
QR certificate + authentication	✓ signed QR, revocation	HSM-held key
Validity tracking, alerts	✓ 30/15/7/1-day reminders	nightly server job
SMS / email updates	🔴 email + SMS wired, keys pending	NIC SMS gateway
Dashboards: pending, inspections	✓ stage, ageing, district	state-wide wallboard
Search, export, print	✓ search, PDF, CSV	bulk audit export
Mobile app for field officers	✓ installable, offline	Android package

TESTING WITH REAL LEGAL METROLOGY OFFICERS (before final build)

Who 5 LMOs + 1 GATC, one district	What 10 real inspections each, on their phones	Measure minutes per inspection, errors, usability score (SUS)	Pass mark SUS ≥ 70, no data-entry blockers
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Feasible now: open-source stack, free tiers cover a district pilot, mirrors today's LMO / GATC process, follows the LM Act 2009 and GATC Rules 2013.

POTENTIAL CHALLENGES AND RISKS → STRATEGIES

No network at site	Offline app; syncs, signs and notifies later
Fake / copied QR	Signed QR + revocation list; serial shown on scan
Rules differ by State	Fees and validity are data per State
Data privacy (DPDP 2023)	Row-level security, minimal public fields
AI mistakes	Advisory only; officer decides; logged
Officer adoption	Field trial first, few-tap phone flow



Complete live demo

Scan for the guided demo: register → apply → officer inspects (try offline) → signed certificate → scan the QR on another phone. Each step ticks itself.

Evaluator password: TulaDemo@2026

tula-legal-metrology.vercel.app/demo



IMPACT AND BENEFITS



POTENTIAL IMPACT ON THE TARGET AUDIENCE

LMOs & GATCs

- Jobs on the phone, even offline
- Pass / fail calculated for them
- Certificate issued on site

Businesses

- Apply and pay without a visit
- SMS / email at every step
- Reminder before a stamp expires

Consumers

- Scan any certificate QR
- Fakes and edits are rejected
- Report a problem in two taps

Government

- Pending by stage, age, district
- Inspections due and overdue
- Full audit trail per instrument

MEASURED ON THE LIVE SYSTEM (30 Sep 2026, scripted end-to-end run on the production database)

2 min 51 s

application filed → signed certificate (system time, 3 people, 16 logged actions)

6 s

certificate issued → signed by the server

6 / 6

forged or edited QR copies rejected (automated tests)

0

paper forms and office visits in the run

PILOT TARGETS vs TODAY'S PAPER BASELINE (baseline measured from the pilot district's register)

≤ 7 days

application → certificate (median)

< 3 s

certificate check by QR scan

100%

due instruments reminded 30 days early

0

paper certificates in the pilot district

BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL, ETC.)

Social

Fair weight at shops, pumps and mandis. Buyers can check.

Economic

Fewer visits and faster renewals for traders and MSMEs.

Environmental

Paperless forms, certificates and records.

Governance

One auditable trail from application to enforcement.

DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK

RESEARCH FINDING	DESIGN DECISION IN TULA
1 Sec. 24 of the Act needs verification and stamping. GATC Rules 2013 add approved test centres.	One workflow for LMO and GATC, routed by district and equipment
2 Fees and validity differ across State Enforcement Rules.	Rules stored as data per State; database isolates each State
3 MPE limits come from OIML R 76 (scales) and R 117 (fuel dispensers).	Pass / fail computed from recorded readings, not typed
4 Field sites (mandis, highways) often have weak networks.	Installable offline app with a sync queue
5 A buyer cannot tell a genuine stamp from a fake.	Signed QR anyone can check, no login, even offline

References & Links

1. The Legal Metrology Act, 2009 · consumeraffairs.gov.in
2. Legal Metrology (General) Rules, 2011
3. LM (Government Approved Test Centre) Rules, 2013
4. State LM (Enforcement) Rules
5. OIML R 76-1 and R 117-1 · oiml.org
6. IT Act 2000, DPDP Act 2023
7. Supabase RLS, Vercel functions docs
8. Google Gemini API docs · ai.google.dev



Scan: a real certificate
 DL/LM/2026/08913, issued and signed on the live system today. Your phone checks the signature.
 Code: github.com/prerak2507/tula-legal-metrology